

MiniBox+ Aging Unit: Clinical & Operational Concerns

Reasons to plan replacement of an end-of-life MiniBox+ PFT system

Per the PulmOne MiniBox+ User Manual (PUO-DOC20001 Rev. 20, §15.9), **the manufacturer-stated expected lifetime of the MiniBox+ is 10 years**. As a unit approaches or exceeds that window, the components below begin to degrade in ways that directly affect test quality, throughput, and patient care. The following issues are commonly encountered with aging units:

Problems encountered with older MiniBox+ units

- **Tablet slowdown & UI lag.** The controlling tablet/PC slows measurably over time — longer boot times, sluggish patient lookup, delayed screen transitions between maneuvers, and occasional freezes mid-test. What used to be a 15-minute PFT can balloon into 25–30 minutes of staff time.
- **Workflow disruptions & lost clinic throughput.** Aging systems increasingly require reboots, re-calibrations mid-session, and repeat maneuvers. Patients are kept waiting, RTs fall behind schedule, and billable studies get pushed to another day. In a busy clinic this is a direct revenue and satisfaction hit.
- **Fast-response occluder valve malfunction → inaccurate LVMS.** The MiniBox+ measures lung volumes using brief airway occlusions performed by an automated fast-response valve (manual spec: **<0.1 sec**). As the valve ages, closure time drifts, seating becomes inconsistent, and occlusions may stick open or closed. The result: **erroneous FRC, RV, and TLC values** — and patients can be **misdiagnosed** with restriction, hyperinflation, or air-trapping they do not actually have (or, worse, a true abnormality can be missed).
- **DLCO gas analyzer drift → unreliable diffusion results.** The infrared CO/CH₄ analyzer has a manufacturer-stated neutral drift spec of **<2% per week**. Older cells drift faster and calibrate out of range, producing DLCO values that fail ATS/ERS quality criteria — again risking misdiagnosis of diffusion impairment.
- **Flow/volume sensor fault → invalid spirometry.** Pneumotach wear and sensor fault (explicitly listed as a replace-device condition in §15.11) cause FEV₁/FVC to drift outside the $\pm 2.5\%$ accuracy spec. Grades drop and repeat testing is required.
- **Pump, valves, and internal electronics nearing end-of-life.** Diaphragm pumps, solenoid valves, capacitors on the main and sensor boards, and cooling fans all have finite service lives. As they fail they produce intermittent errors that are difficult to reproduce on a service call.

PulmOne's own replacement triggers — Manual §15.11, Common Issues:

- **Valve stuck in closed or open position** → "Contact PulmOne support to replace the device."
- **Flow sensor fault** → "Contact PulmOne support to replace the device."

These are not opinions — they are the manufacturer's published replacement conditions.

Bottom line

A MiniBox+ that is at or past its 10-year service life does not fail cleanly — it fails *quietly*, by producing numbers that still look plausible but no longer meet spec. That is the real clinical risk: **patients being diagnosed, treated, or cleared on the basis of degraded data**. Planning a replacement now — rather than waiting for a hard failure — protects patient safety, protects clinic throughput and revenue, and avoids an unplanned downtime event during a full testing schedule.

Sources: PulmOne MiniBox+ User Manual, PUO-DOC20001 Rev. 20 — §3.1 Device Specifications; §15.9 Maintenance Schedule & Lifetime; §15.11 Common Issues. Prepared for field / clinical use.